

$1024 \times 1024 = 10^6$

1 sec $\rightarrow 10^7 - 10^8$ iterations.

$$N \leq 10^6$$

Sorting.

- $O(N)$ ✓
- $O(N^2)$ ✗
- $O(N \log N)$ ✓

$\rightarrow 10^6 \times \log_2(10^6)$
 $\rightarrow 10^6 \times 20$

$$N \leq 10^{10} \rightarrow$$

Binary Search

- $O(N)$ ✗
- $O(\log N)$ ✓
- $O(\sqrt{N})$ ✓
- $O(1)$ ✓

Amazon
MS

Adobe

1. Given an array of 1's & 0's. We can replace any 1 of the 0's with a 1. Return the count of max. consecutive 1's in the array.

Ex:-

	0	1	2	3	4	5	6	7
	1	1	0	1	1	0	1	1

size of 5

5

2

→ ans = 5

Q:-

1	1	0	1	1	0	1	1	1
---	---	---	---	---	---	---	---	---

5

3

→ ans = 6

Ex:-

0	1	1	1	0	1	1	0	1	1	0
---	---	---	---	---	---	---	---	---	---	---

2 + 2 + 1 = 5

0 + 3 + 1 = 4

3 + 2 + 1 = 6

2 + 0 + 1 = 3

6

ans = 0;

for every 0 in the array :-

→ Count no. of consecutive 1's in the left (l)

→ Count no. of consecutive 1's in the right (r)

→ If $(l + r + 1 > \text{ans})$
 $\{ \text{ans} = l + r + 1 \}$

return ans;

0 0 0 0 0
↓
 $l = 0; r = 0$
 $l + r + 1 \geq \text{ans}; \text{ans} = 1;$
↪ ans = 1;

Edge Case:

1 1 1 1 1

↪ ans is length of array.

↪ Outer loop

$$\begin{aligned}
 & \text{for } (i \rightarrow 0 \text{ to } N) \{ \\
 & \quad \text{if } (A[i] == 0) \\
 & \quad \{ \text{traverse left; } \} \quad \text{inner loops} \\
 & \quad \text{traverse right; } \} \\
 & \}
 \end{aligned}$$

$$\text{for } (j = i-1; j \geq 0; j--)$$

$$as = 0 + 3 + 1 = 4; \quad 7.$$

Observation:

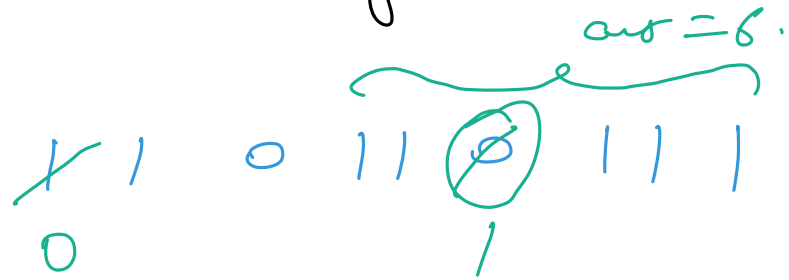
At max, every element 0 being accessed 3 times.

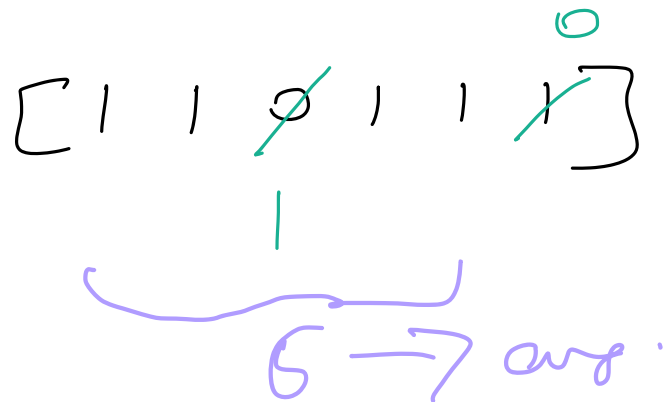
Total iterations $\leq 3N$.

$$T.C = O(N)$$

Amey or / Direct-i / ms / Adobe ...

2. Given an array of 1's & 0's,
we can swap one of the
0's with a 1. Return the
count of max consecutive 1's
in the array.

Ex:- 

Q.: 

Step 1 :-

Calculate the total no. of
1's in my array.
 $\hookrightarrow C.$

For every 0 in the array:

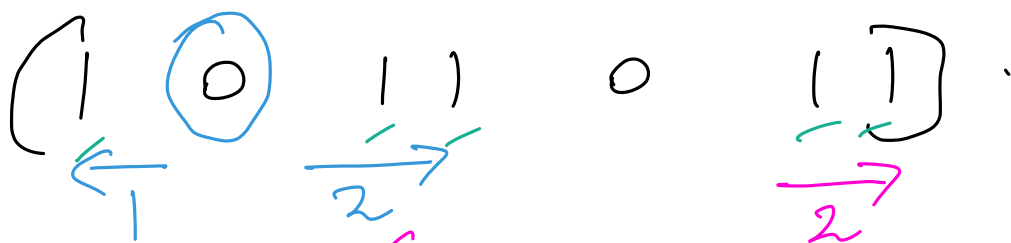
$\rightarrow$ Count consecutive 1's in the
left (l)

$\rightarrow$ Count consecutive 1's
in the right (r)

$$\text{count} = \begin{cases} (l+r+1) & \text{if } (l+r) < C. \\ (l+r) \text{ OR } C. & \text{if } (l+r) == C. \end{cases}$$

if ($\text{count} > \text{ans}$) { $\text{ans} = \text{count};$ }

$$C = 5$$



$$l + r = 3 < 5$$

$$; ans = 4; 5;$$

$$l + r = 2 + 2 = 4 < 5$$

Case 2r



$$l + r = 2 + 2 = 4 == 4$$

$$Count = 2 + 2 = 4$$

$$\begin{aligned} ans &= \max(ans, count) \\ &= \max(0, 4) \\ &= 4 \end{aligned}$$

$$T.C \rightarrow O(N).$$

Amey on
VMWare.

3. No. of triplets
Given an array. Count the
no. of triplets i, j, k
such that
 $i < j < k$
& $A[i] < A[j] < A[k]$.

Ex: $(\overset{0}{2}, \overset{1}{6}, \overset{2}{9}, \overset{3}{4}, \overset{4}{10})$.

i	j	k	$A[i]$	$A[j]$	$A[k]$
0	1	2	2	6	9
0	1	4	2	6	10
0	3	4	2	4	10
0	2	4	2	9	10
1	2	4	6	9	10

$$\text{ans} = \underline{\underline{6}}$$

Idea Iterate over all possible triplets.

count = 0;

for (i = 0 → N-3)

{ for (j = i+1 to N-2)

T.C → $O(N^3)$ { for (k = j+1 to N-1)

S.C → $O(1)$ { if ($A[i] < A[j]$ & $A[j] < A[k]$)
count++;

}
}
}
return count;

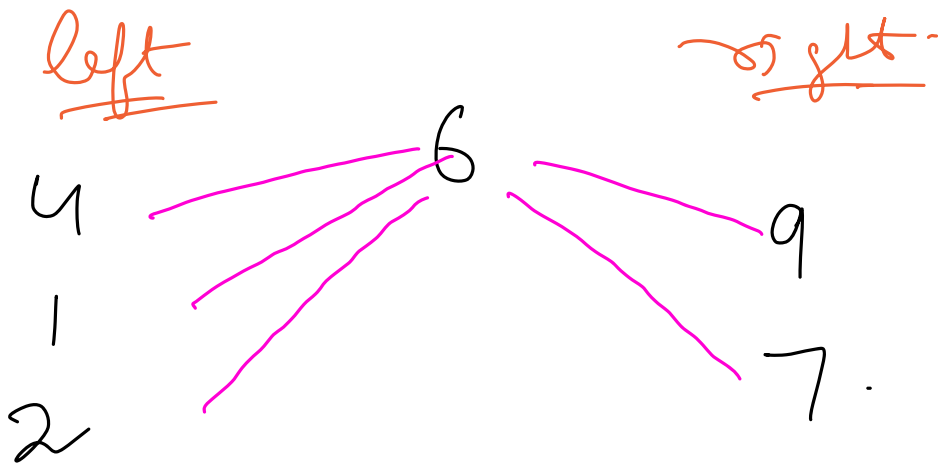
A = 0 1 2 3 4 5 6.
 4 1 2 6 9 7 2

On how many triplets, the

element 6 is in the middle.

$$i < \textcircled{1} < k$$

3



$$\underline{i < j < k.}$$

4	6	9
4	6	7
1	6	9
1	6	7
2	6	9
2	6	7

Total
3 x 2 = 6

No. of triplets in which
A[i] is in middle.

= No. of elem $< A[i]$ on left of i .

~~X~~

No. of elem $> A[i]$ on right of i .

1 2 6
1 2 9
1 2 7
4

1 2 6 9 7.

0 0 1 3 4 4.

3 4 3 2 0 0
0 0 3 6 0 0

Ans = 9.

for every element $A[i]$:

- iterate from index
0 to $i-1$

T.C $\rightarrow O(N^2)$

& count the elements
smaller than $A[i] \rightarrow l$

$\rightarrow$ iterate from index
 $i+1$ to $N-1$ &
count the elements
greater than $A[i] \rightarrow r$.

$ans = ans + (l * r);$

return ans;

$N = 100;$

$i = 0;$

0

99.

$i = 1.$

1

98.

$i = 2.$

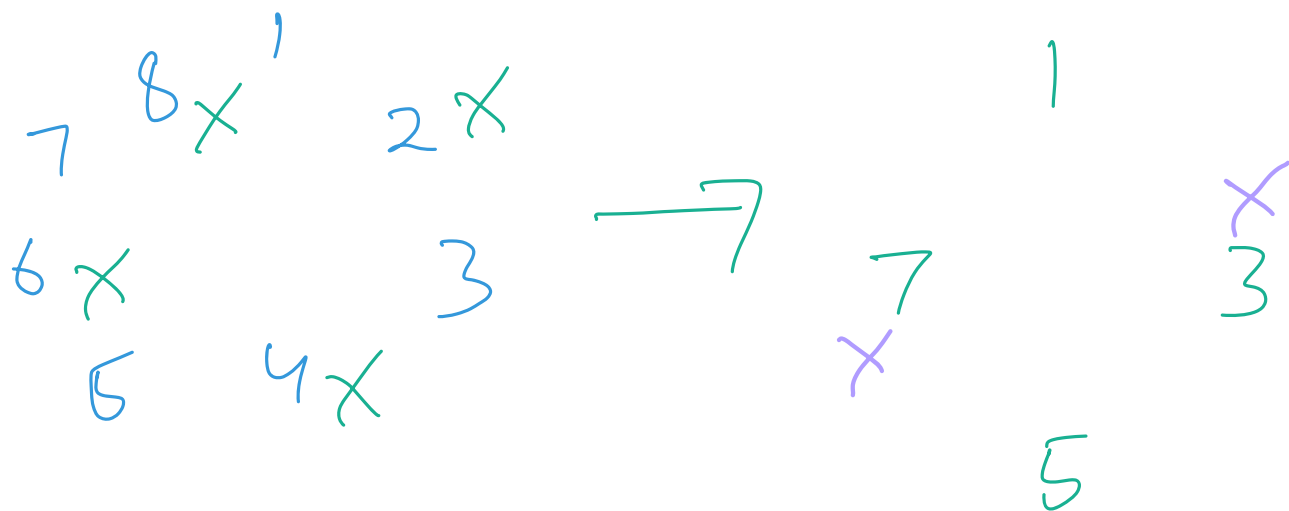
2

Q 7.

For each i ; my inner loop will run $(N-1)$ times

Q. Josephus Problem.

There are N people who are standing in a circle. Every time, an alternate person is evicted from that circle. You have to tell me what is the index of the final person which is remaining.

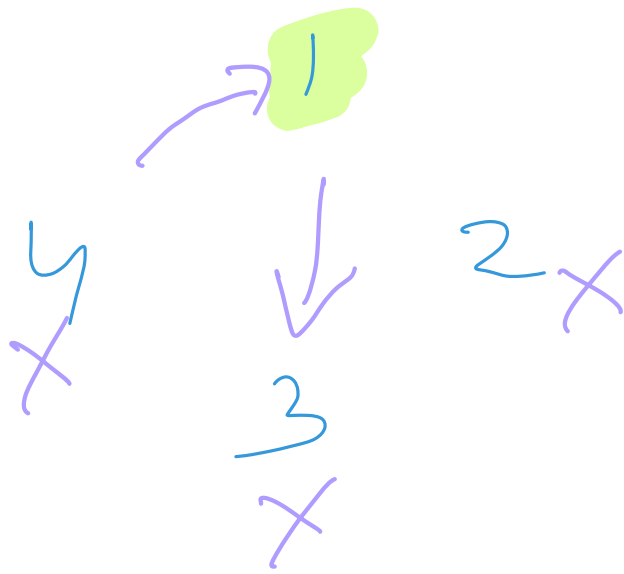


Q1:-

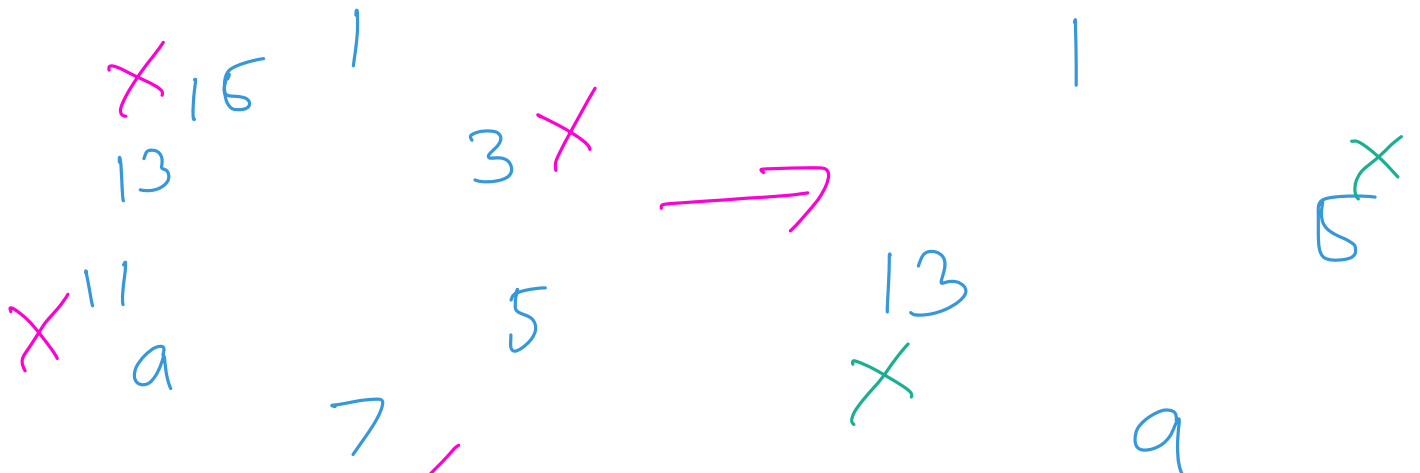
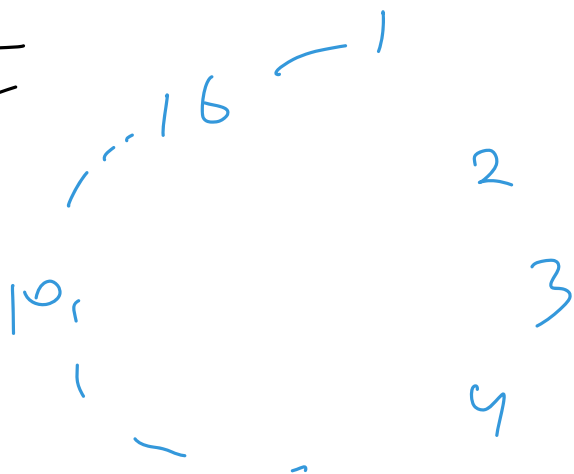
1

2 x

ar



ar



x



1

9
x

Q 1

How about $N=6$;

1 x

x 6

6

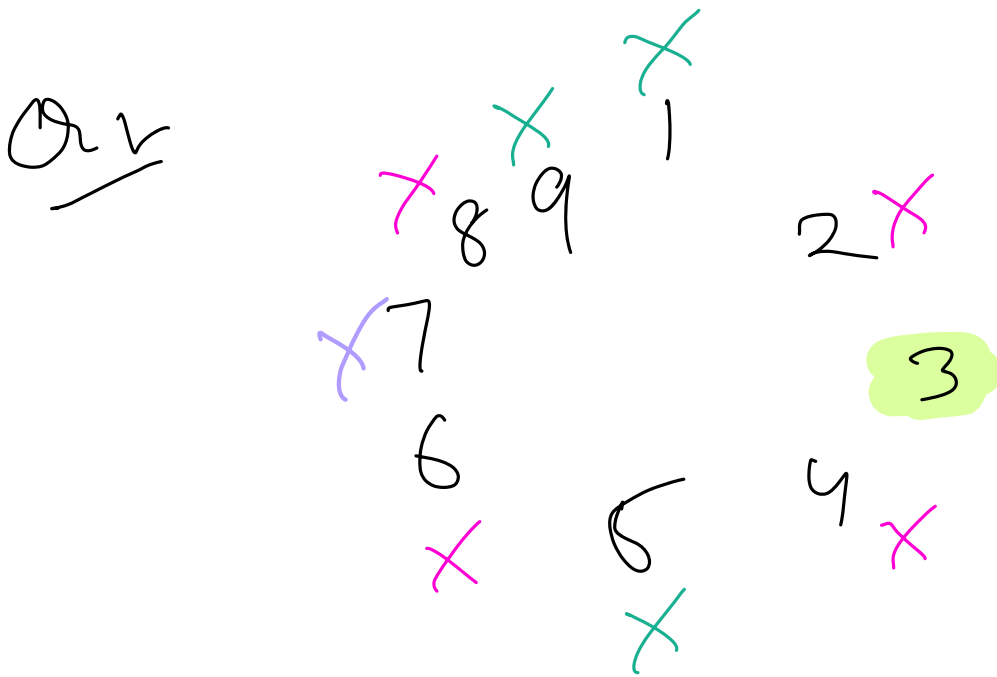
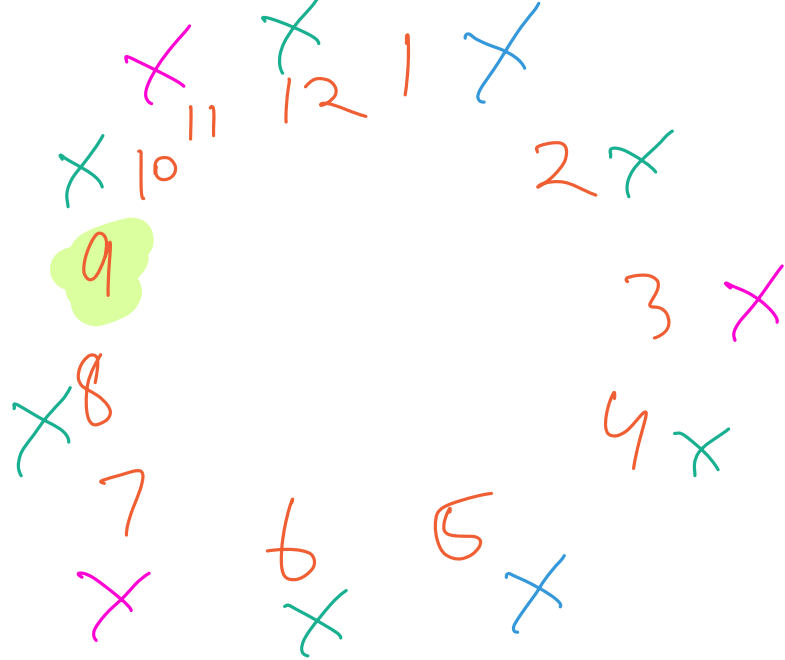
2 x

3 x

4 x

Q 2

12.



Obs $\rightarrow$ 1.

All the powers of 2
will have as ~~as~~ 1.

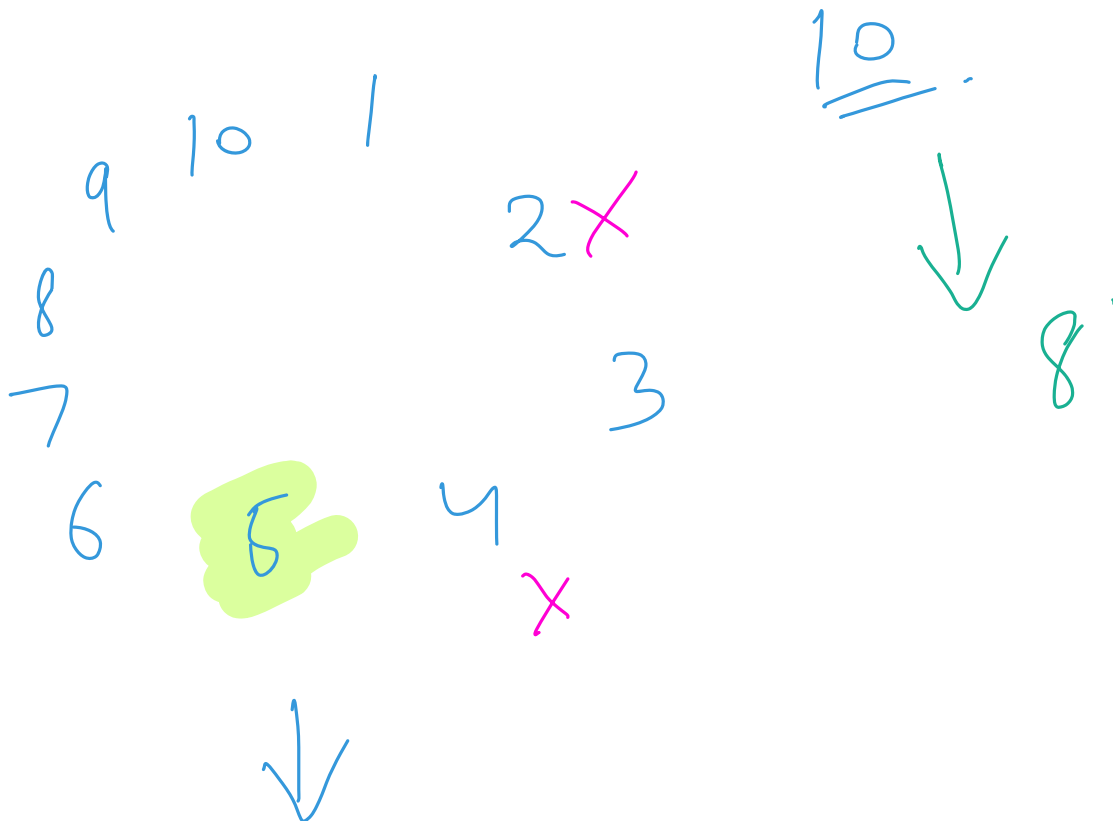
2, 4, 8, 16.

The starting element in the
of power of 2 will
give me the
answer.

Can I reduce my N to
the nearest power of 2
& then realise the
starting position

Exr

$$N = 10$$



3 5
 1 6
 10 7
 9 8

12.

12 $\rightarrow$ 8.

11 12 1
 10

2~~x~~

$12 - 8$
 $= \underline{4}.$

9

3

$1 + 2(4)$
 $= \underline{9}.$

4~~x~~

8~~x~~

7

6~~x~~

5

Q 2

11

11 $\rightarrow$ 8

$$y = \underline{3}$$

9 10 11 1

2 $\times$

8

3

7

4 $\times$

6

5

$\times$

$$1 + \underline{2 \times 3}$$

$\downarrow$

$$\underline{N - 2^x}$$

Approach.

Step 1: Calculate the greatest power of 2 $\leq N$.

N .

2^x .

$$N - 2^x = y$$

Step 2 !: Cal. the starting
point which is

$$1 + 2xy.$$

Find power of 2 nearest to N .
100.

2, 4, 8, 16, 32, 64,
128.

Announcements:

Conts 1 — Reorthpt 2.



Next July 1st

Sat & Sunday. 2nd.

Contes 2 $\rightarrow$ Reattempt 1.

$\hookrightarrow$ It is active now
up till 10:30pm
tomorrow.

July 1 $\rightarrow$ July 9.

Replacement Instructor.

Strong Marking 1 & Marking 2.

$N = \underline{\underline{11}}$
 $\downarrow$
8
6x
7
8x
9
10x
11
1
2
3
4
5

$$5 + \underline{\underline{2 \times 3}} \\ = 76 = 11$$

Doublet :

